

METAGENOME PROFILER

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- SARS-CoV-2 emerged December 2019
 - primary host: horseshoe bat (*Rhinolophidae*), Yunnan, China [1]
- Zhou et al. [2] identified novel CoV strain (RmYN02) in *Rhinolophus malayanus* feces via shotgun metagenomics
- Outbreaks will continue as humans encroach on animal spaces
- Environmental monitoring is critical for pandemic readiness
- Monitoring viral communities enables prediction of future spread



[3]

Data

Reads

- Short-read shotgun metatranscriptomic sequencing of feces from *Rhinolophus malayanus* bats [2]

Reference Genome

- *Rhinolophus ferrumequinum* shotgun WGS assembly
 - Closest one I could find to *R. malayanus*

Pipeline

1. Host read removal:

K=31 kmer index of host genome; reads with >50% kmer match removed

2. Quality filtering:

reads with avg Phred < 20 removed

3. Viral classification:

Kraken2 LCA exact k-mer matching against viral database [4]

4. Abundance analysis:

species/genus/family counts ÷ total classified

5. Read edit distance analysis:

pairwise Levenshtein distance → MDS + PERMANOVA

- Started with 39,257,492 reads
- full 28-chromosome run estimated at 2,770,000,000,000 hours (Figure 1)
 - ↳ index built against only the 2nd and 3rd chromosome scaffolds
- Host filter (chr. 2 & 3) removed 147,140 reads → **39,110,352**

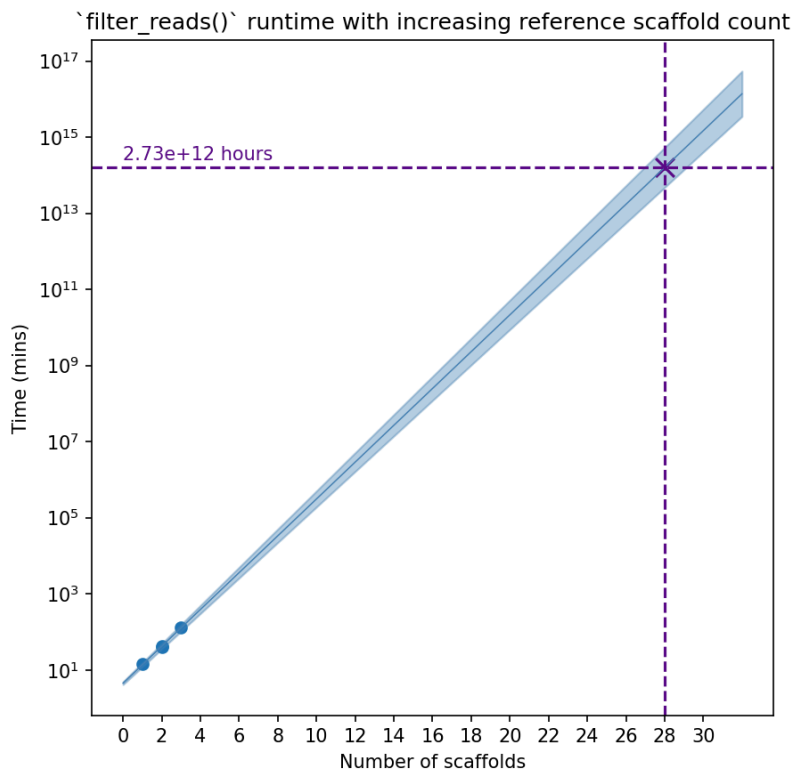


Figure 1: Projected runtime vs. scaffold count.

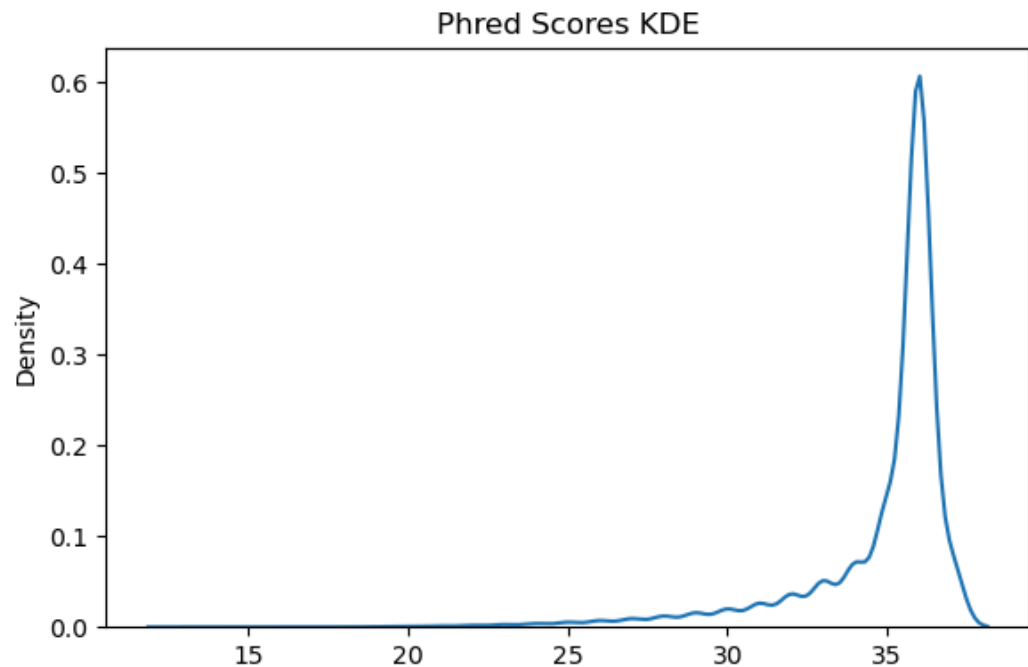


Figure 2: Read Phred score distribution.
Density peaks around 36–37; very little density below 20.

- Only **22,647 reads (0.06%)** fell below **Phred threshold of 20**
 - **high overall sequencing quality.**

- **286,284** of 39,257,492 reads classified against Kraken2 viral database.
- #Species : #Genus/Family ratio of **38:1** means reads resolve well to species level.

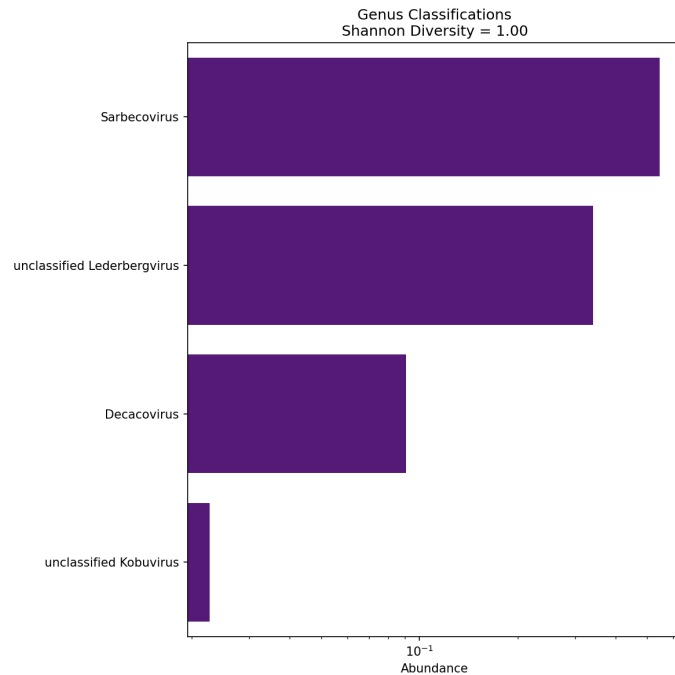


Figure 3: Genus log-abundances.

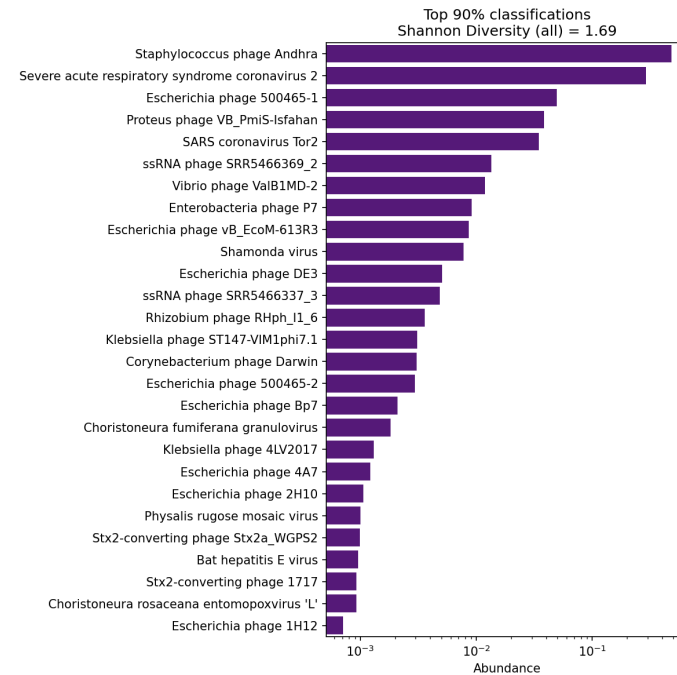


Figure 4: Top 90% Species log-abundances.

- Pairwise Levenshtein distances on stratified 10,000-read sample
- PERMANOVA
 - pseudo-F = 133.5, **p = .001**
 - Species membership is a strong predictor of read-sequence similarity
- MDS reveals clear clustering by species

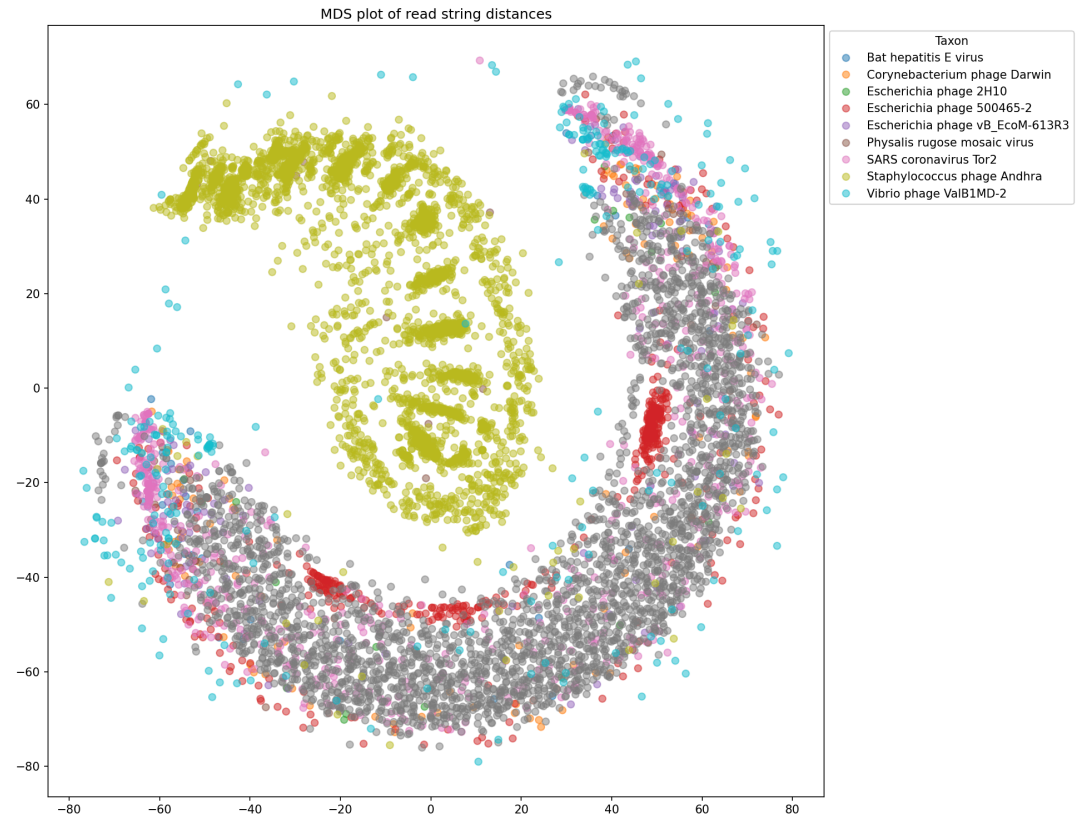


Figure 5: MDS of read Levenshtein distances, colored by species.

- SARS-CoV-2 and SARS CoV Tor2 rank among top 5 most abundant species (Figure 4)
 - replicates Zhou et al.: *R. malayanus* hosts CoV [2]
 - Supports *Rhinolophidae* are CoV reservoirs [1]
- Virome dominated by bacteriophages
- Edit distance analysis provides complementary confirmation
 - reads from the same species are more similar to each other than to reads from other species (Figure 5)
- Shotgun metagenomics is a valid tool for monitoring pathogen zoonosis

REFERENCES

- [1] Y. Han *et al.*, “Panoramic analysis of coronaviruses carried by representative bat species in Southern China to better understand the coronavirus sphere,” *Nat. Commun.*, vol. 14, no. 1, p. 5537, Sept. 2023.
- [2] H. Zhou *et al.*, “A novel bat Coronavirus closely related to SARS-CoV-2 contains natural insertions at the S1/S2 cleavage site of the spike protein,” *Curr. Biol.*, vol. 30, no. 11, pp. 2196–2203.e3, June 2020.
- [3] “Rhinolophus affinis.” [Online]. Available: https://commons.wikimedia.org/wiki/File:Rhinolophus_affinis_86903750.jpg
- [4] D. E. Wood, J. Lu, and B. Langmead, “Improved metagenomic analysis with Kraken 2,” *Genome Biol.*, vol. 20, no. 1, p. 257, Nov. 2019.

https://github.com/canderson318/metagenome_profiler